STAT 201A Homework 2

Show all work and justify every step from definitions, axioms, or results proved in class. You may discuss the problems with classmates, but the solutions you submit must be written by you. Pdf files of typed or hand-written-and-scan answers are both acceptable.

Covers: counting, conditional probability and Bayes' rule, independence, random variables, cdfs, pmfs, and pdfs (Weeks 2–3, C&B §§1.2.3–1.6).

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1. (Exercise 1.21) A closet contains n pairs of shoes. If $2r$ shoes are chosen at random $(2r < n)$, what is the probability that there will be no matching pair in the sample?

2. (Exercise 1.33) Suppose that 5% of men and .25% of women are color-blind. A person is chosen at random and that person is color-blind. What is the probability that the person is male? (Assume males and females to be in equal numbers.)

3. (Exercise 1.38) Prove each of the following statements. (Assume that any conditioning event has positive probability.)

- **(a)** If $P(B) = 1$, then $P(A \mid B) = P(A)$ for any A.
- **(b)** If $A \subset B$, then $P(B \mid A) = 1$ and $P(A \mid B) = P(A)/P(B)$.
- **(c)** If A and B are mutually exclusive, then

$$P(A \mid A \cup B) = \frac{P(A)}{P(A) + P(B)}.$$

- **(d)** $P(A \cap B \cap C) = P(A \mid B \cap C)P(B \mid C)P(C)$.

4. (Exercise 1.39) A pair of events A and B cannot be simultaneously *mutually exclusive* and *independent*. Prove that if $P(A) > 0$ and $P(B) > 0$, then:

- **(a)** If A and B are mutually exclusive, they cannot be independent.
- **(b)** If A and B are independent, they cannot be mutually exclusive.

5. Two medical tests. A disease has prevalence 1%. A screening test is positive with probability 0.95 for a person with the disease and with probability 0.10 for a person without it. A randomly chosen person takes the test twice. Let D = "the person has the disease" and T_i = "test i is positive," $i = 1, 2$. Assume the two results are independent **given** the disease status: $P(T_1 \cap T_2 \mid D) = P(T_1 \mid D)P(T_2 \mid D)$, and the same with D^c in place of D.

- **(a)** Find $P(D \mid T_1)$.

- **(b)** Find $P(D \mid T_1 \cap T_2)$.
- **(c)** Show that you get the same answer as (b) by applying Bayes' rule to test 2 alone, using your answer to (a) as the "prior" probability of D. Explain why this works.
- **(d)** Are T_1 and T_2 independent? Compute $P(T_1 \cap T_2)$ and $P(T_1)P(T_2)$, and explain in words why "independent given D" does not imply "independent."

6. (Exercise 1.47) Prove that the following functions are cdfs.

- **(a)** $\frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$, $x \in (-\infty, \infty)$
- **(b)** $e^{-e^{-x}}$, $x \in (-\infty, \infty)$
- **(c)** the function defined in (1.5.6)

For each, state whether the corresponding random variable is continuous, discrete, or neither (Definition 1.5.7).

7. A cdf with jumps. Let X have cdf

$$F(x) = \begin{cases} 0 & x < 0 \\ x/4 & 0 \le x < 1 \\ 1/2 & 1 \le x < 2 \\ 1 - \frac{1}{4}e^{-(x-2)} & x \ge 2. \end{cases}$$

Write $F(a^-) = \lim_{x \uparrow a} F(x)$.

- **(a)** For any random variable with cdf F and any $a < b$, show that $P(a < X \le b) = F(b) - F(a)$ and $P(X = a) = F(a) - F(a^-)$. *Hint:* $\{X < a\} = \bigcup_{n=1}^{\infty} \{X \le a - \frac{1}{n}\}$. You may use the fact that if $A_1 \subset A_2 \subset \cdots$, then $P\big(\bigcup_n A_n\big) = \lim_{n\to\infty} P(A_n)$.
- **(b)** Check that F satisfies the three conditions of Theorem 1.5.3. Is X continuous, discrete, or neither?
- **(c)** Find $P(X = 1)$, $P(X = 2)$, $P(\frac{1}{2} < X < 2)$, $P(X > 3)$, and $P(X \ge 1 \mid X \le 2)$.
- **(d)** Describe a two-stage experiment that produces a random variable with cdf F: first flip a fair coin, then, depending on the result, draw X from a discrete distribution or from a continuous one. Specify both distributions.

8. (Exercise 1.55) An electronic device has lifetime denoted by T. The device has value $V = 5$ if it fails before time $t = 3$; otherwise, it has value $V = 2T$. Find the cdf of V, if T has pdf

$$f_T(t) = \frac{1}{1.5} e^{-t/(1.5)}, \qquad t > 0.$$

9. A pmf and a pdf with the same tail.

- **(a)** Let $f_X(k) = \dfrac{c}{k(k+1)}$ for $k = 1, 2, \ldots$, and 0 otherwise. Find the constant c that makes f_X a pmf. *Hint:* $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$.

- **(b)** Find the cdf $F_X(x)$ for **every** real x, and sketch it. Find $P(X > n)$ for $n = 1, 2, \ldots$
- **(c)** Let $f_Y(y) = \dfrac{c}{y^2}$ for $y > 1$, and 0 otherwise. Find c so that f_Y is a pdf, and find $F_Y(y)$ and $P(Y \geq n)$.
- **(d)** Show that $\lfloor Y \rfloor$, the integer part of Y, has the same distribution as X. *Hint:* find $P(\lfloor Y \rfloor = k)$ by writing the event $\{\lfloor Y \rfloor = k\}$ in terms of Y.
- **(e)** A pmf always satisfies $f_X(x) \leq 1$. Explain why. Does a pdf have to satisfy $f_Y(y) \leq 1$? Prove it or give a counterexample.

Problem 1:

There are n pairs, $2n$ shoes total, and we draw $2r$ of them with $2r < n$

Total outcomes:

Any set of $2r$ shoes is equally likely: $\binom{2n}{2r}$

Outcomes with no matching pair:

Every shoe has to come from a different pair

- Pick which $2r$ pairs the shoes come from: $\binom{n}{2r}$ ways
- From each chosen pair, pick the left or right shoe: 2^{2r} ways

$$P(\text{no matching pair}) = \frac{\binom{n}{2r} 2^{2r}}{\binom{2n}{2r}}$$

Problem 2:

Let M = male, C = color-blind.

We know $P(M) = P(M^c) = 0.5$, $P(C|M) = 0.05$, $P(C|M^c) = 0.0025$

By Baye's rule, we have

$$P(M|C) = \frac{P(C|M)P(M)}{P(C|M)P(M) + P(C|M^c)P(M^c)} = \frac{(0.05)(0.5)}{(0.05)(0.5) + (0.0025)(0.05)} = \frac{20}{21} \approx 0.952$$

Problem 3

Definition 1.3.2. If A and B are events in S, and $P(B) > 0$, then the **conditional probability of A given B**, written $P(A \mid B)$, is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{1.3.1}$$

Interpretation. B becomes the new sample space: $P(B \mid B) = 1$, and each event is "rescaled" by how much of it lies in B. In particular, if A and B are disjoint, then $P(A \mid B) = P(B \mid A) = 0$.

a) If $P(B) = 1$, then $P(B^c) = 0$. Since $A \cap B^c \subset B^c$, monotonicity gives $0 \leq P(A \cap B^c) \leq P(B^c) = 0$

$\Rightarrow P(A \cap B^c) = 0$

Split A into disjoint pieces $A \cap B$ and $A \cap B^c$: $P(A) = P(A \cap B) + P(A \cap B^c) = P(A \cap B)$

$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{1} = P(A)$

b) If $A \subset B$, then $A \cap B = A$

$$\Rightarrow P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A)}{P(A)} = 1 \quad , \quad P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)}$$

c)

Since $A \subset A \cup B$, we have $A \cap (A \cup B) = A$

Since A and B are mutually exclusive, additivity gives $P(A \cup B) = P(A) + P(B)$

$$\Rightarrow P(A|A \cup B) = \frac{P(A \cap (A \cup B))}{P(A \cup B)} = \frac{P(A)}{P(A)+P(B)}$$

d)

$$P(A|B \cap C) \cdot P(B|C) \cdot P(C) = \frac{P(A \cap B \cap C)}{P(B \cap C)} \cdot \frac{P(B \cap C)}{P(C)} \cdot P(C) = P(A \cap B \cap C)$$

Problem 4:

Let $P(A) > 0$ and $P(B) > 0$, we have $P(A)\,P(B) > 0$

a) If A and B are mutually exclusive, then $A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$

However $P(A)\,P(B) > 0$, so $P(A \cap B) \neq P(A)\,P(B)$

They are not independent

b) If A and B are independent, then $P(A \cap B) = P(A)\,P(B) > 0$

An events with positive probability can't be empty, so $A \cap B \neq \emptyset$

They are not mutually exclusive

Problem 5

Given $P(D) = 0.01$, $P(T_i|D) = 0.95$, $P(T_i|D^c) = 0.10$

a) Law of total probability:

Bayes: $P(T_1) = P(T_1|D)\,P(D) + P(T_1|D^c)\,P(D^c) = (0.95)(0.01) + (0.10)(0.99) = 0.1085$

$$\Rightarrow P(D|T_1) = \frac{P(T_1|D)\,P(D)}{P(T_1)} = \frac{(0.95)(0.01)}{0.1085} \approx 0.0876$$

b)

By conditional independence:

$$P(T_1 \cap T_2 | D) = 0.95^2 = 0.9025 \qquad P(T_1 \cap T_2 | D^c) = 0.10^2 = 0.01$$

Total probability:

$$P(T_1 \cap T_2) = P(T_1 \cap T_2|D)\,P(D) + P(T_1 \cap T_2|D^c)\,P(D^c) = (0.9025)(0.01) + (0.01)(0.99)$$
$$= 0.018925$$

Bayes:

$$P(D|T_1 \cap T_2) = \frac{P(T_1 \cap T_2|D)\,P(D)}{P(T_1 \cap T_2)} = \frac{(0.9025)(0.01)}{0.018925} = \frac{361}{757} \approx 0.477$$

c)

Let $p' = P(D|T_1) = \frac{19}{217}$ as the new prior and update on T_2 only:

$$\frac{0.95\,p'}{0.95p' + 0.10(1-p')} = \frac{0.95(19)}{0.95(19) + 0.10(198)} = \frac{18.05}{37.85} = \frac{361}{757} \quad \text{(same answer as b)}$$

Explain: The real Bayes update after T_1 is

$$P(D|T_1 \cap T_2) = \frac{P(T_2|D \cap T_1)\,P(D|T_1)}{P(T_2|D \cap T_1)\,P(D|T_1) + P(T_2|D^c \cap T_1)\,P(D^c|T_1)}$$

Conditional independence lets us drop T_1 from the likelihoods:

$$P(T_2 \mid D \cap T_1) = \frac{P(T_1 \cap T_2 \mid D)}{P(T_1 \mid D)} = \frac{P(T_1 \mid D)\, P(T_2 \mid D)}{P(T_1 \mid D)} = P(T_2 \mid D)$$

Same process again with D^c

Thus once you know disease status, the first result tells you nothing extra about the second, and updating one test at a time gives the same posterior as updating on both at once.

d)

$$P(T_1 \cap T_2) = 0.018925 \qquad P(T_1)\,P(T_2) = 0.1085^2 \approx 0.01177$$

These results are not equal, so T_1 and T_2 are not independent.

In words: both tests depend on the same hidden thing, the disease status. If I don't know that status, a positive first test makes it more likely the person is sick, which in turn makes a positive second test more likely. The tests are linked through their common cause. Conditioning on D removes that link, which is why they're independent given D but not conditionally

Problem 6

Theorem 1.5.3. The function $F(x)$ is a cdf if and only if the following three conditions hold:

- **a.** $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.
- **b.** $F(x)$ is a nondecreasing function of x.
- **c.** $F(x)$ is right-continuous; that is, for every number x_0, $\lim_{x\downarrow x_0} F(x) = F(x_0)$.

a) $F(x) = \frac{1}{2} + \frac{1}{\pi}\tan^{-1}(x)$, $x \in (-\infty, \infty)$

i) $\tan^{-1}(x) \to -\pi/2$ as $x \to -\infty$, so $F \to \frac{1}{2} - \frac{1}{2} = 0$

$\tan^{-1}(x) \to \pi/2$ as $x \to \infty$, so $F \to 1$

ii) $F'(x) = \frac{1}{\pi(1+x^2)} > 0$, so F is increasing

iii) F is continuous everywhere, so it's right-continuous

Since F is continuous, X is continuous $\Rightarrow$ this function is cdf

b) $F(x) = e^{-e^{-x}}$

i) As $x \to -\infty$, $e^{-x} \to \infty \Rightarrow F \to e^{-\infty} = 0$

As $x \to \infty$, $e^{-x} \to 0 \Rightarrow F \to e^{0} = 1$

ii) $F'(x) = e^{-x} e^{-e^{-x}} > 0$

iii) It is composition of continuous function, so it is continuous

Thus, X is continuous, and this function is cdf

c) **Example 1.5.6 (Cdf with jumps).** A cdf can mix continuous pieces and jumps. For some ϵ with $1 > \epsilon > 0$, let

$$F_Y(y) = \begin{cases} \dfrac{1-\epsilon}{1+e^{-y}} & \text{if } y < 0 \\ \epsilon + \dfrac{1-\epsilon}{1+e^{-y}} & \text{if } y \geq 0. \end{cases} \tag{1.5.6}$$

F_Y has a jump of height ϵ at $y = 0$ and is otherwise continuous (Exercise 1.47). Think of a gauge whose reading could be any real number but that sometimes sticks at 0, with ϵ the probability that it sticks.

i) As $y \to -\infty$, $1+e^{-y} \to \infty \Rightarrow F \to 0$

As $y \to \infty$, $F \to \varepsilon + (1-\varepsilon) = 1$

ii) $\dfrac{1}{1+e^{-y}}$ is increasing, so each piece is increasing

At $y=0$, the left limit is $\dfrac{1-\varepsilon}{2}$ and $F(0) = \dfrac{\varepsilon + (1-\varepsilon)}{2}$, which is bigger

$\Rightarrow$ F never decreases across the break

iii) Each piece is continuous. At 0, $F(0)$ is given by the right piece, so $\lim_{y \downarrow 0} F(y) = F(0)$

F jumps by ε at 0, so it isn't continuous, and it strictly increases between jumps, so it isn't a step function

X is neither

This function is cdf

Problem 7

a) Since $\{X \leq a\} \subset \{X \leq b\}$ and $\{a < X \leq b\} = \{X \leq b\} \setminus \{X \leq a\}$,

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a)$$

The sets $A_n = \{X \leq a - \frac{1}{n}\}$ are increasing, $A_1 \subset A_2 \subset \ldots$, and their union is $\{X < a\}$

$\Rightarrow$ by continuity of probability, $P(X < a) = \lim_{X \to \infty} F(a - \frac{1}{n}) = F(a^-)$

The last step holds because F is non decreasing and bounded, so the left limit exists and equals the limit along any sequence increasing to a.

$$P(X=a) = P(X \le a) - P(X<a) = F(a) - F(a^-)$$

b)

i) $F(x) = 0$ for all $x<0$, so the limit at $-\infty$ is 0

As $x \to \infty$, $1 - \frac{1}{4}e^{-(x-2)} \longrightarrow 1$

ii) Each piece is non decreasing: $x/4$ increases, $1/2$ is flat, $1 - \frac{1}{4}e^{-(x-2)}$ increases

Check the breaks:

- At 0: left limit 0, $F(0) = 0$
- At 1: left limit $\frac{1}{4}$, $F(1) = \frac{1}{2}$. Jump up of $\frac{1}{4}$
- At 2: left limit $\frac{1}{2}$, $F(2) = 1 - \frac{1}{4} = \frac{3}{4}$. Jump up of $\frac{1}{4}$

So F never decreases

iii) Each piece is continuous, and every interval is closed on the left

$\Rightarrow F(a)$ equals the right limit at each break

F has jumps at 1 and 2, so it isn't continuous. It increases smoothly on $(0,1)$ and $(2,\infty)$, so it isn't a step function.

X is neither

c)

$$P(X=1) = F(1) - F(1^-) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$P(X=2) = F(2) - F(2^-) = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$$

$$P\left(\frac{1}{2} < X < 2\right) = P\left(\frac{1}{2} < X \le 2\right) - P(X=2) = F(2^-) - F\left(\frac{1}{2}\right) = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$

$$P(X>3) = 1 - F(3) = \frac{1}{4}e^{-1} \approx 0.092$$

$$P(X \ge 1 \mid X \le 2) = \frac{P(1 \le X \le 2)}{P(X \le 2)} = \frac{F(2) - F(1^-)}{F(2)} = \frac{\frac{3}{4} - \frac{1}{4}}{\frac{3}{4}} = \frac{1/2}{3/4} = \frac{2}{3}$$

d) The jumps carry total mass $\frac{1}{4}+\frac{1}{4}=\frac{1}{2}$, and the smooth parts carry the other $\frac{1}{2}$

density $\frac{1}{4}$ on $(0,1)$ gives $\frac{1}{4}$ and density $\frac{1}{4}e^{-(x-2)}$ on $(2,\infty)$ gives $\frac{1}{4}$

So the coin is fair

Flip a fair coin, we get:

- Heads: draw X from discrete distribution $P(X=1)=P(X=2)=\frac{1}{2}$
- Tails: draw X from the continuous density

$$g(x)=\begin{cases}\frac{1}{2} & 0<x<1\\ \frac{1}{2}e^{-(x-2)} & x>2\\ 0 & \text{otw}\end{cases}$$

Check that $F=\frac{1}{2}F_{disc}+\frac{1}{2}F_{cont}$:

- $0\le x<1$: $\frac{1}{2}(0)+\frac{1}{2}\left(\frac{x}{2}\right)=\frac{x}{4}$
- $1\le x<2$: $\frac{1}{2}\left(\frac{1}{2}\right)+\frac{1}{2}\left(\frac{1}{2}\right)=\frac{1}{2}$
- $x\ge 2$: $\frac{1}{2}(1)+\frac{1}{2}\left[\frac{1}{2}+\frac{1}{2}\left(1-e^{-(x-2)}\right)\right]=1-\frac{1}{4}e^{-(x-2)}$

Problem 8

T is exponential with mean 1.5, so $P(T \le t) = 1 - e^{-t/1.5}$ for $t > 0$

$V = 5$ if $T < 3$, and $V = 2T \ge 6$ if $T \ge 3$. So V can only be 5 or something in $[6, \infty)$

- $v < 5$: V is never below 5, so $F_V(v) = 0$
- $5 \le v < 6$: $V \le v$ only when $V = 5$

  For example, $T < 3$:

$$F_V(v) = P(T < 3) = 1 - e^{-3/1.5} = 1 - e^{-2}$$

- $v \ge 6$: $V \le v$ happens when $T < 3$ or when $T \ge 3$ and $2T \le v$:

$$F_V(v) = P(T < 3) + P(3 \le T \le \tfrac{v}{2}) = P(T \le \tfrac{v}{2}) = 1 - e^{-(\frac{v}{2})/1.5} = 1 - e^{-v/3}$$

$$F_V(v) = \begin{cases} 0 & v < 5 \\ 1 - e^{-2} & 5 \le v < 6 \\ 1 - e^{-v/3} & v \ge 6 \end{cases}$$

Problem 9

a) The partial sums telescope:

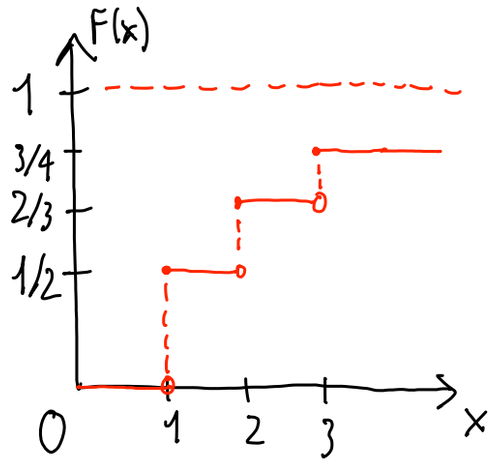
$$\sum_{k=1}^{N} \frac{c}{k(k+1)} = c \sum_{k=1}^{N} \left(\frac{1}{k} - \frac{1}{k+1} \right) = c \left(1 - \frac{1}{N+1} \right) \xrightarrow{N \to \infty} c$$

For a pmf the total has to be 1, so $c = 1$ ($f_X(k) > 0$ for all k, so non negativity holds)

b) For $x < 1$, $F_X(x) = 0$. For $k \le x < k+1$ with $k \ge 1$, the same telescoping gives

$$F_X(x) = \sum_{j=1}^{k} \frac{1}{j(j+1)} = 1 - \frac{1}{k+1}$$

$$\Rightarrow F_X(x) = 1 - \frac{1}{\lfloor x \rfloor + 1} \quad \text{for } x \ge 1$$



Tail: $P(X>n) = 1 - F_X(n) = \frac{1}{n+1}$

c)

$$\int_1^\infty \frac{c}{y^2}\,dy = c\left[-\frac{1}{y}\right]_1^\infty = c \Rightarrow c = 1$$

Then $F_Y(y) = 0$ for $y \le 1$, and for $y > 1$

$$F_Y(y) = \int_1^y \frac{1}{t^2}\,dt = 1 - \frac{1}{y}$$

Since Y is continuous, $P(Y = n) = 0$, so for $n \ge 1$

$$P(Y \ge n) = 1 - F_Y(n) = \frac{1}{n}$$

d)

$Y > 1$, so $\lfloor Y \rfloor$ takes values $1, 2, \ldots$

For $k \ge 1$, $\{\lfloor Y \rfloor = k\} = \{k \le Y < k+1\}$,

$$P(\lfloor Y \rfloor = k) = F_Y(k+1) - F_Y(k) = \frac{1}{k} - \frac{1}{k+1} = \frac{1}{k(k+1)} = f_X(k)$$

Same support, same probabilities, so $\lfloor Y \rfloor$ has the same distribution as X

e)

A pmf is a probability, $f_X(x) = P(X=x)$, and $\{X=x\} \subset S \Rightarrow f_X(x) \leq P(S) = 1$

A pdf does not satisfy $f_Y(y) \leq 1$, because a density value isn't a probability, only its integral over a set is

Counter example:

Uniform $(0, \frac{1}{2})$ has $f(y) = 2$ for $0 < y < \frac{1}{2}$, which is nonnegative and integrates to $2 \cdot \frac{1}{2} = 1$

However, it equals to 2 everywhere on its support

A density can even be unbounded like $f(y) = \frac{1}{2\sqrt{y}}$ on $(0,1)$